$$a^{\ln n} = n^{\ln a} = \frac{1}{n^{-\ln a}} = \frac{1}{n^{\left(\ln \frac{1}{a}\right)}} \quad \ln\frac{1}{a} = p.$$

$$\sum \frac{1}{n^p}$$

conv. if $p > 1$

div. if $p \leq 1$.

⇒ HW self-study : next ex.

Seminar 4

① a) $\displaystyle\sum_{n\geq 1} \frac{(-1)^{n+1}}{\sqrt{n(n+1)}}$

$\boxed{\text{Leibniz Test: If } x_n \searrow 0, \text{ then } \sum (-1)^n \cdot x^n \text{ is convergent}}$.

$$\frac{1}{\sqrt{n(n+1)}} \searrow 0 \checkmark \xrightarrow[\text{Test}]{\text{Leibniz}} \text{convergent}.$$

A.C.? $\displaystyle\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{\sqrt{n(n+1)}} \right| = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+1)}} =$

$\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^p}$ convergent if $p > 1$, divergent if $p < 1$

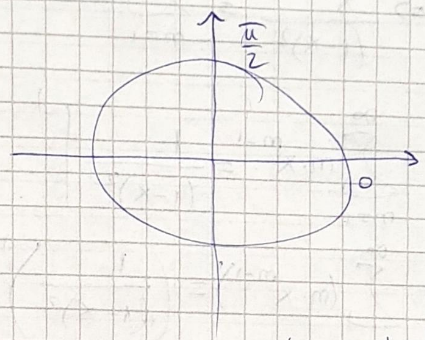
$$\frac{1}{n\sqrt{n+1}} = \sqrt{\frac{1}{n^2+1}} \simeq \frac{1}{n}$$

$$\lim_n \frac{\frac{1}{n\sqrt{n+1}}}{\frac{1}{n}} = 1. \Rightarrow \sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+1)}} \text{ has the same nature as}$$

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty \text{ (divergent)}$$

⇒ not absolutely convergent.

② b) $\sum_{n=1}^{\infty} (-1)^n \underbrace{\sin \frac{1}{n}}_{x_n}$



$\left.\begin{array}{l} \sin \frac{1}{n} \to 0 \\ \sin \text{ increasing on } \left[0, \frac{\pi}{2}\right] \end{array}\right\}$

$\Rightarrow \sin \frac{1}{n} \searrow 0.$

$\left((1-x)^{-1}\right)' = -(1-x)^{-2} \cdot (-1)$

$\xRightarrow{\text{Leibniz test}} x_n$ convergent.

A.C.? $\sum_{n=1}^{\infty} \left| (-1)^n \cdot \sin \frac{1}{n} \right| = \sum_{n=1}^{\infty} \left| \sin \frac{1}{n} \right| \xlongequal{\sin \frac{1}{n} > 0} \sum_{n=1}^{\infty} \sin \frac{1}{n}$ "like" $\sum \frac{1}{n}$

$\frac{\sin \frac{1}{n}}{\frac{1}{n}} \to 1 \in (0, \infty) \Rightarrow$ they have the same nature.

③ c) $\sum_{n=1}^{\infty} \left| \frac{\sin n}{n^2} \right| \leq \sum_{n=1}^{\infty} \left| \frac{1}{n^2} \right| = \sum_{n=1}^{\infty} \frac{1}{n^{2}} < \infty \Rightarrow$ convergent. ($2 > 1$)

$-1 \leq \sin n \leq 1 \Rightarrow |\sin n| \leq 1.$

$\Rightarrow$ absolutely convergent $\Rightarrow$ convergent.

② Prove by differiatiating the geometric series that for $|x| < 1$, we have the following:

$$\sum_{n=1}^{\infty} n \cdot x^n = \frac{x}{(1-x)^2}, \quad \sum_{n=2}^{\infty} n \cdot (n-1) \cdot x^n = \frac{2x^2}{(1-x)^3}$$

The geom. series: $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \Big|'$, $|x| < 1.$

$\to \sum_{n=0}^{\infty} (-x)^n = \sum_{n=0}^{\infty} (-1)^n \cdot x^n = \frac{1}{1+x}$

$\Rightarrow \left(\sum_{n=0}^{\infty} x^n \right)' = \sum_{n=0}^{\infty} (x^n)'$ because it is absolutely convergent.

$\Leftrightarrow \left(\frac{1}{1-x} \right)' = \sum_{n=1}^{\infty} n \cdot x^{n-1} \Leftrightarrow \frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n \cdot x^{n-1} \Big| \cdot x$

$$\Rightarrow \frac{x}{(1-x)^2} = \sum_{n=1}^{\infty} n \cdot x^n$$

$$\cancel{\left((1-x)^{-2}\right)' = +2 \cdot (1-x)}$$

$$\sum_{n=1}^{\infty} n \cdot x^{n-1} = \frac{1}{(1-x)^2} \Big| '$$

$$\left((1-x)^{-2}\right)' = -2 \cdot \frac{(1-x)'}{(1-x)^3}$$

$$\Rightarrow \sum_{n=1}^{\infty} (n \cdot x^{n-1})' = \left(\frac{1}{(1-x)^2}\right)'$$

$$\Rightarrow \sum_{n=2}^{\infty} n(n-1) \cdot x^{n-2} = -\frac{-2(1-x)'}{(1-x)^3} = \frac{2}{(1-x)^3} \Big| \cdot x^2$$

$$\Rightarrow \sum_{n=2}^{\infty} n(n-1) \cdot x^n = \frac{2x^2}{(1-x)^3}$$

③ Prove by integrating the geometric series that for $|x| < 1$,

(1) $\sum_{n=1}^{\infty} \frac{x^n}{n} = -\ln \cancel{\frac{1}{x}} (1-x)$

(2) $\sum_{n=1}^{\infty} (-1)^{n+1} \cdot \frac{x^n}{n} = \ln(1+x).$

$$\int x^n \, dx = \frac{x^{n+1}}{n+1}$$

$$\int_0^x t^n \, dt = \frac{t^{n+1}}{n+1} \Big|_0^x = \frac{x^{n+1}}{n+1}$$

$$\sum_{n=1}^{\infty} t^n = \frac{1}{1-t} \Big| \int_0^x \quad , (\forall) t \in (-1,1)$$

$$, (\forall) x \in (-1,1)$$

$$\int_0^x \frac{-(1-t)'}{1-t} \, dt$$

$$-\ln(1-t) \Big|_0^x$$

$$(\Leftrightarrow) \int_0^x \left(\sum_{n=1}^{\infty} t^n\right) dt = \int_0^x \frac{1}{1-t} \, dt$$

$0 < t < x < 1.$

$$(\Leftrightarrow) \sum_{n=0}^{\infty} \int_0^x t^n \, dt = \int_0^x \frac{1}{1-t} \, dt$$

$$(\Leftrightarrow) \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} = \int_0^x \frac{1}{1-t} \, dt \Rightarrow$$

$\Rightarrow$ ~~$-\ln(1-t)\Big|_0^x =$~~

$\Leftrightarrow \displaystyle\sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} = -\ln(1-t)\Big|_0^x$

$\Leftrightarrow \displaystyle\sum_{n=1}^{\infty} \frac{x^n}{n} = -\ln(1-x)$

$x = 1 \to -\ln 0 \to -(-\infty) = \infty$

$\Leftrightarrow -\ln(1-x) = -\ln(1-x)$, true, $\underline{(\forall)\, x \in (-1,1)}$

To obtain (2) $\Rightarrow$ replace $x := -x$

$\underline{(1)} \Rightarrow \displaystyle\sum_{n=1}^{\infty} \frac{(-x)^n}{n} = -\ln(1+x) \ \Big|\cdot(-1)$

$\Rightarrow \displaystyle\sum_{n=1}^{\infty} (-1)^{n+1} \cdot \frac{x^n}{n} = \ln(1+x)$, $(\forall)\, x \in (-1,1)$.

$x = 1 \Rightarrow$ alternating harmonic series $\to \ln 2$.

(4) $\displaystyle\sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{2n+1}}{2n+1}$

$\boxed{\left(\frac{1}{1+x^2}\right)' = \operatorname{arctg} x}$

$\boxed{\left(\frac{1}{x}\right)' = \ln x}$

$\displaystyle\sum_{n=0}^{\infty} t^n = \frac{1}{1-t}$, $\underline{(\forall)\, t \in (-1,1)}$

$t = -t$

$\displaystyle\sum_{n=0}^{\infty} (-1)^n t^n = \frac{1}{1+t}$, $\underline{(\forall)\, t \in (-1,1)}$

$t \mapsto t^2$ ("replace")

$t = t^2$

$\displaystyle\sum_{n=0}^{\infty} (-1)^n \cdot t^{2n} = \frac{1}{1+t^2}$, $(\forall)\, t \in (-1,1) \ \Big|\ \int_0^x dt$, $\underline{x \in (-1,1)}$.

$\Rightarrow \displaystyle\sum_{n=0}^{\infty} \int_0^x (-1)^n \cdot t^{2n}\, dt = \int_0^x \frac{1}{1+t^2}\, dt$

$\Rightarrow \displaystyle\sum_{n=0}^{\infty} (-1)^n \cdot \int_0^x t^{2n}\, dt = \operatorname{arctg} t \Big|_0^x$

$\Rightarrow \displaystyle\sum_{n=0}^{\infty} (-1)^n \cdot \frac{t^{2n+1}}{2n+1}\Big|_0^x = \operatorname{arctg} x - \operatorname{arctg} 0$

$\Rightarrow \displaystyle\sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{2n+1}}{2n+1} = \operatorname{arctg} x$ (arctg $\mapsto$ arctan)

For $x=1$: $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \arctan 1 = \frac{\pi}{4} \Rightarrow$ convergent (Leibniz test)

by continuity

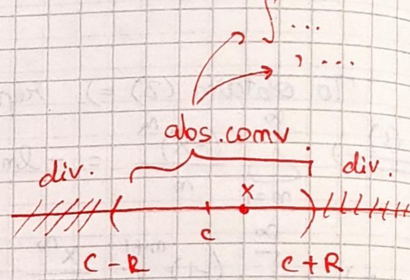
(we extended the int. from $(-1,1)$ to sth. bigger

why?

For $x=-1$: $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{2n+1} = -\frac{\pi}{4}$ $\qquad (-1)^n \cdot (-1)^{2n+1} = (-1)^n \cdot (-1) = (-1)^{n+1}$

⑤ Radius of convergence

$\displaystyle\sum_{n=0}^{\infty} a_n (x-c)^n$ power series structure.



when?

$L = \lim_n \sqrt[n]{|a_n|} = \lim_n \left| \frac{a_{n+1}}{a_n} \right|$

$R = \frac{1}{L}$ „radius of convergence".

$\exists$ ↳ the series is • absolutely conv. for $(\forall)\, x \in (c-R, c+R)$
$|x-c| < R$.

• divergent for $(\forall)\, |x-c| > R$.

Convergence set

$(c-R, c+R)$, check $c-R$, $c+R$

$(c-R, c+R) \cup \underbrace{\{c-R, c+R\}}_{\text{to check}}$.

a) $\displaystyle\sum_{n=0}^{\infty} \frac{x^n}{2^n}$, $c=0$, $a_n = \frac{1}{2^n}$.

$L = \lim_n \left| \frac{a_{n+1}}{a_n} \right| = \lim_n \left| \frac{\frac{1}{2^{n+1}}}{\frac{1}{2^n}} \right| = \frac{1}{2}$.

$R = \frac{1}{L} = \frac{1}{\frac{1}{2}} = 2$ radius of convergence.

$\Rightarrow$ the series is abs. convergent, $(\forall)\, x \in (-2,2)$

$x=2 \Rightarrow \sum_{n=0}^{\infty} \frac{2^n}{2^n} = \sum 1 = \infty$ divergent

$x=-2 \Rightarrow \sum_{n=0}^{\infty} \frac{(-2)^n}{2^n} = \sum (-1)^n$ divergent

$\Rightarrow C = (-2, 2)$

c) $\sum_{n \geq 1} \frac{(x-2)^n}{(n+1) \cdot 3^n}$ $c = 2$, $a_n = \frac{1}{(n+1) \cdot 3^n}$, $\cancel{L = \lim_n}$

$L = \lim_n \left| \frac{a_{n+1}}{a_n} \right| = \lim_n \left| \frac{\frac{1}{(n+2) 3^{n+1}}}{\frac{1}{(n+1) \cdot 3^n}} \right| = \frac{1}{3}$

$\Rightarrow R = \frac{1}{L} = \frac{1}{3}$ abs. convergent, $(\forall) x \in (2 - \overset{3}{\frac{1}{3}}, 2 + \overset{3}{\frac{1}{3}})$

$x \in (-1, 5)$

$x = 5: \sum_{n \geq 1} \frac{1}{n+1} = \infty$

$x = -1: \sum_{n \geq 1} \frac{(-1)^n}{n+1}$ convergent (Leibniz test)

$\Rightarrow$ Conv. set: $[-1, 5)$

d) $\sum_{n=0}^{\infty} n! \cdot x^n$, $c = 0$, $a_n = n!$

$L = \lim_n \frac{(n+1)!}{n!} = n+1 = \infty$

$\Rightarrow R = \frac{1}{L} = \frac{1}{\infty} = 0$

($c = 0$, $R = 0 \Rightarrow (0-0, 0-0) = \{0\}$)

Convergence set: $\{0\} = C$

e) $\sum_{n \geq 1} \frac{(x-1)^n}{n^p}$, $\underline{p > 0}$. $c = 1$, $a_n = \frac{1}{n^p}$

$\frac{1}{n} : \frac{1}{n^p}$

$L = \lim_n \frac{n^p}{(n+1)^p} = \lim_n \left(\frac{n}{n+1} \right)^p = \cancel{\lim_n} = 1^p = 1.$

$\cancel{\frac{\frac{1}{n^{p+1}}}{\frac{1}{n^p}} = \frac{n^p}{n^{p+1}} = \frac{1}{n}}$.

$\Rightarrow R = \frac{1}{1} = 1$; $\cancel{C}$

$\Leftrightarrow$ abs. convergent, $(\forall) x \in (0, 2)$

$x=2 \Rightarrow \sum_{n\geq 1} \frac{1}{n^{p}} \Rightarrow$ convergent if $p>1$, $p>1$

$x=0 \Rightarrow \sum_{n\geq 1} \frac{(-1)^n}{n^p} \overset{(*)}{\Rightarrow}$ convergent

$\Rightarrow C = \begin{cases} [0,2] & , p>1 \\ [0,2) & , p\leq 1 \end{cases}$

$\frac{1}{n^p} \searrow 0 \xrightarrow{\text{(Leibniz test)}}$ convergent. $(*)$

f) $\sum_{n=1}^{\infty} \frac{(-1)^n \cdot x^n}{\sqrt{n}}$, $c=0$, $a_n = \frac{(-1)^n}{\sqrt{n}}$

$L = \lim_n \left| \frac{a_{n+1}}{a_n} \right| = \lim_n \left(\frac{(-1)^{n+1}}{\sqrt{n+1}} \cdot \frac{\sqrt{n}}{(-1)^n} \right) = \dots = 1$

$\Rightarrow R = \frac{1}{L} = 1 \Rightarrow$ abs. convergent, $(\forall) x \in (-1,1)$

$x=+1$: $\sum \frac{(-1)^n}{\sqrt{n}}$ conv.

$\frac{1}{\sqrt{n}} \searrow 0 \xrightarrow[\text{Test}]{\text{Leibniz}}$ convergent

$x=-1$: $\sum \frac{(-1)^n \cdot (-1)^n}{\sqrt{n}} = \sum \frac{1}{\sqrt{n}} = \sum \frac{1}{n^{\frac{1}{2}}} \overset{\infty}{=}$ divergent, $\frac{1}{2}<1$

$\Rightarrow C = (-1, 1]$